

Hallucination in Large Language Models: A Survey of Detection, Evaluation, and Mitigation

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Abstract

Large language models (LLMs) support a wide range of language tasks, but hallucinations remain a major obstacle to their reliable use. Their outputs may contradict established facts or introduce information unsupported by the available evidence. This survey presents an overview of hallucination in text-based LLMs, bringing together its definitions, detection methods, evaluation practices, and mitigation strategies. We begin by distinguishing world factuality from source faithfulness and illustrating common error types. We then examine detection approaches based on external evidence, sampling consistency, and internal model signals, alongside representative benchmarks and evaluation methods. We further discuss potential causes of hallucination and review interventions involving training, retrieval, decoding, and response revision. Through these comparisons, we examine the strengths and limitations of existing approaches and the trade-offs among factual accuracy, informative coverage, and computational cost. Finally, we discuss open challenges in evidence reliability, independent verification, and generalization, outlining directions for more reliable language generation. A companion repository of relevant literature and supporting materials is available at <https://github.com/lixincheng-xcl/LLM-Hallucination-Survey>.

1 Introduction

A fluent answer from a large language model (LLM) is not necessarily justified. In question answering (QA), a model may repeat a misconception; in summarization, it may add a plausible detail absent from the source. Both are called hallucinations, although their correctness criteria differ: source-unsupported content can be factually correct (Lin et al., 2022; Cao et al., 2022). Collapsing these settings obscures what a system improves.

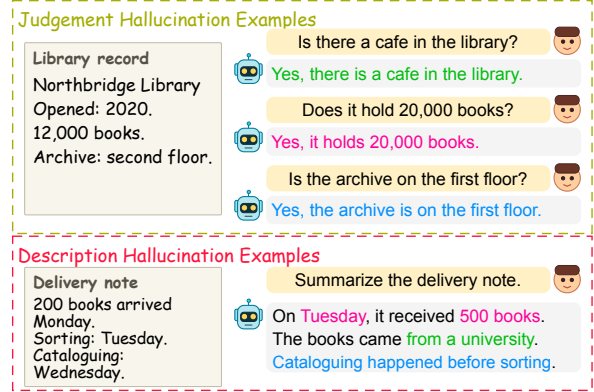


Figure 1: Illustrative hallucination examples in source-grounded question answering and summarization. Green, pink, and blue denote unsupported additions, attribute errors, and relation errors, respectively. Unsupported content is not necessarily false.

In Figure 1, the library record supports 12,000 books and a second-floor archive. The cafe assertion lacks support; the changed count and location contradict the record. The delivery summary also invents a university source and reverses events. Verification must specify the claim and its reference evidence.

Building on the broad factuality review of Wang et al. (2024), we connect each method’s evidence and access assumptions to the evaluation target it can support. This comparison distinguishes better correction from gains caused by withholding information or changing the verifier. Three questions organize the synthesis: **RQ1**, what reference frame defines an error; **RQ2**, what evidence and access enable detection; and **RQ3**, when does mitigation improve useful generation?

Scope and organization. This narrative survey covers 2022–2026 ACL-family work on text QA, long-form generation, summarization, dialogue, and retrieval-augmented generation (RAG), with mechanism-defining studies from other venues. Selection prioritized direct factuality/faithfulness rel-



Figure 2: The structure of this survey. Green boxes list representative studies for each branch.

evance, distinct mechanisms, and benchmark diversity. Canonical publications were preferred, retaining evidence that challenges common assumptions. Figure 2 maps concepts (§2), detection (§3), evaluation (§4), mitigation (§5), and analysis (§6). Appendix A details selection. Multimodal perception and general reasoning are excluded.

2 Concepts and Scope

2.1 Reference frames

World factuality concerns agreement with externally established facts; *source faithfulness* concerns support by supplied evidence. Neither entails the other: a response can repeat a false source faithfully or add a true but unlicensed detail (Cao *et al.*, 2022; Adlakha *et al.*, 2024). BEGIN evaluates attribution in grounded dialogue under this distinction (Dziri *et al.*, 2022b).

Source-based labels should distinguish *supported*, *contradicted*, and *not established*, retaining the last separately from world falsity. World verification additionally needs a specified evidence source, date, and unresolved-claim policy. These are operational choices, not a universal definition. HalluLens revisits definitions and benchmark construction (Bang *et al.*, 2025); comparisons must map study-specific labels before comparing scores.

2.2 Error relations and development

Entity, attribute, and relation errors are orthogonal to reference frames: a wrong date can contradict either a document or a world fact. Annotation units also differ. ANAH provides sentence-level analysis, while HaluEval 2.0 extends empirical factuality analysis across domains (Ji *et al.*, 2024; Li *et al.*, 2024).

Research moved from distinct task-specific correctness targets in 2022 toward atomic verification and black-box self-checking in 2023 (Min *et al.*, 2023; Manakul *et al.*, 2023), followed by contextual monitoring and factuality alignment. Recent studies scrutinize judge reliability, information coverage, and temporal validity (Godbole and Jia, 2025; Samarinas *et al.*, 2025; Jiang *et al.*, 2026).

Three axes organize the comparison: reference frame, detection evidence, and intervention point. Benchmarks evaluate generated content or detectors; detectors can guide decoding. Fine-tuning and long-context studies motivate failures in supervision, knowledge access, and evidence use (Gekhman *et al.*, 2024; Liu *et al.*, 2024c). Across axes, we compare evidence access, generator signals, supervision, and cost, separating decision criteria from deployment requirements.

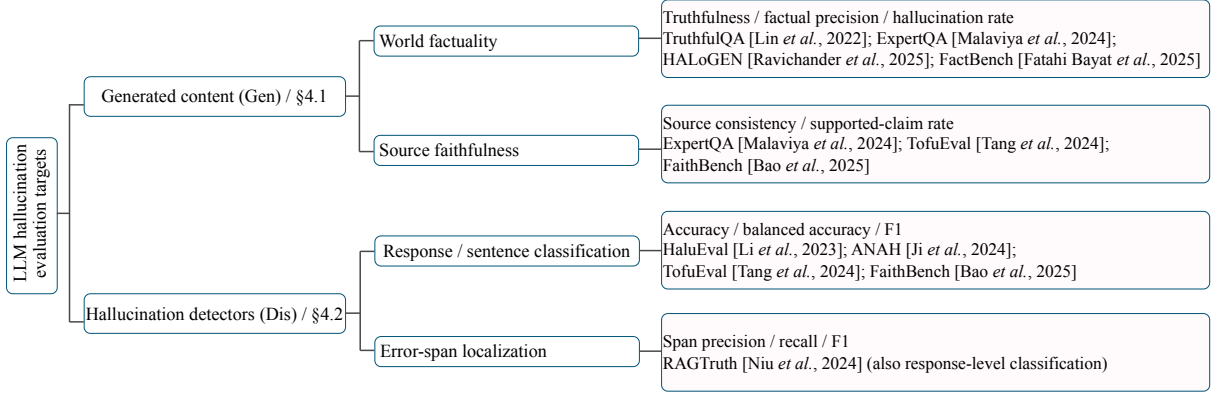


Figure 3: Taxonomy of hallucination evaluation methods for LLMs. The two branches distinguish generated-content evaluation from error-label prediction.

3 Detection: Evidence and Access

Detection estimates errors under a specified reference; Figure 2 groups methods by evidence and model access.

3.1 External evidence

Evidence-based methods compare outputs with supplied or retrieved knowledge. QAFactEval asks questions about summary content and checks answer recovery from the source; question generation and answerability affect its decisions (Fabbri et al., 2022). AlignScore learns alignment across heterogeneous tasks and aggregates source-chunk/output-sentence scores (Zha et al., 2023). Both use trained auxiliaries; neither guarantees correct verification.

FActScore decomposes long answers into atomic facts and estimates their supported fraction relative to a designated source (Min et al., 2023). VeriScore instead extracts verifiable claims, using surrounding context to resolve references (Song et al., 2024). Extraction changes the score denominator; missing support differs from contradiction. PrefixNLI extends entailment to incomplete prefixes, enabling feedback during decoding (Harary et al., 2026).

3.2 Sampling and consistency

SelfCheckGPT scores a response sentence against additional responses to the same prompt (Manakul et al., 2023). Its NLI and prompting variants aggregate contradiction or non-support judgments across samples. This requires repeated generation rather than target-model probabilities; agreement can still reflect a repeated misconception.

Semantic uncertainty instead clusters sampled answers by mutual entailment and estimates entropy over meanings, discounting paraphrase varia-

tion (Kuhn et al., 2023; Farquhar et al., 2024). The weighted estimator needs sequence probabilities; a discrete approximation uses cluster frequencies. SelfCheckGPT localizes inconsistency in a given response, whereas semantic entropy measures dispersion among answer meanings, targeting unstable confabulations. Both depend on sampling and comparison quality; stable false beliefs need not trigger either signal.

3.3 Internal signals

FOCUS combines token probabilities with attention-based uncertainty propagation from an accessible proxy model (Zhang et al., 2023b). The target generator can remain black-box; the proxy must expose both signals. Hidden-state classifiers learn truth labels from activations (Azaria and Mitchell, 2023). Lookback Lens uses context-versus-generation attention features with a supervised linear classifier (Chuang et al., 2024a). Probes require representative training labels.

A separable activation pattern need not represent truth: associated falsehoods can resemble factual recall in controlled experiments (Cheang et al., 2026). Probes therefore need evaluation across natural error types and generators.

Source alignment cannot establish source truth; sampling and probes supply no independent factual evidence. “Black-box” and “reference-free” differ because access to generators, proxies, and verifiers must be specified separately.

4 Benchmarks and Evaluation

Figure 3 separates generation evaluation from detector evaluation. World factuality and source faithfulness are reference frames for both; detector sub-branches distinguish scoring units. Table 1 sizes

Table 1: Hallucination evaluation benchmarks for LLMs. ‘Dis’, ‘Gen’, and ‘MC’ denote error-label discrimination, generated-content evaluation, and multiple choice, respectively; P and R denote precision and recall. Sizes count the stated units. Appendix B details the evaluation protocols.

Benchmark	Setting	Size	Focus	Metrics
TruthfulQA (Lin et al., 2022)	Gen/MC	817 questions (2022)	World factuality	Gen: truthfulness; informativeness; MC1 accuracy; MC2 true-answer mass
HaluEval (Li et al., 2023)	Dis	35,000 samples	Mixed-task errors	Recognition accuracy
ANAH (Ji et al., 2024)	Dis	~12k sentences	QA support; contradiction	Type accuracy; F1
RAGTruth (Niu et al., 2024)	Dis	17,790 responses	RAG faithfulness	Response/span P, R, F1
ExpertQA (Malaviya et al., 2024)	Gen	2,177 questions	Correctness; attribution	Human factuality/support
TofuEval (Tang et al., 2024)	Dis/Gen	1,479 summaries*	Dialogue summaries	Balanced accuracy; error rate
HALoGEN (Ravichander et al., 2025)	Gen	10,923 prompts	Domain factuality	Atomic hallucination rate
FaithBench (Bao et al., 2025)	Dis/Gen	750 summaries	Summary faithfulness	Balanced accuracy; macro F1; conditional error rate
FactBench (Fatahi Bayat et al., 2025)	Gen	1,000 prompts	Verifiable claims	Hallucination score; precision

* Paper-reported retained count; the source arithmetic is inconsistent. HALoGEN’s count includes programming. FaithBench’s selected cases are not a prevalence sample. Dis/Gen indicates distinct possible uses, not a shared metric.

count dataset units, not difficulty.

4.1 Generation evaluation

TruthfulQA evaluates misconception-oriented answers for truthfulness and informativeness, with a separate multiple-choice setting (Lin et al., 2022). ExpertQA distinguishes correctness from attribution through expert questions and claim-level judgments (Malaviya et al., 2024). HALoGEN uses domain-specific verifiers; FactBench combines user-derived prompts with web verification (Ravichander et al., 2025; Fatahi Bayat et al., 2025). Their scores are not interchangeable because evidence and verifier scopes differ.

TofuEval studies topic-focused dialogue summaries, where factuality must coexist with relevance and completeness (Tang et al., 2024). Precision needs coverage and refusal reporting, with explicit empty/unverifiable-answer policies.

Aggregation also changes the target. For $n > 0$ extracted claims and s supported claims, VeriScore combines $P = s/n$ with $R_K = \min(s/K, 1)$ as $F_1@K = 2PR_K/(P + R_K)$ for $s > 0$, and zero otherwise (Song et al., 2024). Its domain-specific $K > 0$ is the median extracted-claim count across evaluated responses. Unlike FActScore’s basic support fraction, this rewards supported-claim quantity as well as precision. However, R_K is not coverage of the information required by a question: extra supported details can raise it while leaving important aspects unanswered. Comparisons should fix

K and separately audit extraction, verification, and task-relevant coverage. FactBench’s undecidable units additionally require an explicit aggregation policy (Fatahi Bayat et al., 2025).

4.2 Detector evaluation

HaluEval mixes generated task-specific examples with human-annotated general queries; recognition accuracy need not transfer to unselected errors (Li et al., 2023). ANAH annotates sentence-level types, evidence, and corrections; RAGTruth adds naturally generated RAG responses with error spans (Ji et al., 2024; Niu et al., 2024). Response classification and span localization are different tasks: finding one error does not locate every unsupported claim.

Table 1’s corpora do not supply every detector input. On fixed RAGTruth outputs, alignment can use source–response pairs (Niu et al., 2024). Self-CheckGPT additionally needs prompt-conditioned samples from a declared generator; Lookback Lens needs attention features and probe-training labels. Specify feature models and prefixes; regenerating answers changes the evaluated outputs. Sentence-to-token conversion adds localization assumptions. TofuEval’s sentence and summary decisions also require separate assessment (Tang et al., 2024).

FaithBench selects cases by detector disagreement, so its error frequency is conditional on selection (Bao et al., 2025). TRIVIA+ tests long contexts and noisy supervision (Chen et al., 2026).

Accuracy weights classes by prevalence; balanced accuracy weights them equally. Precision/recall require a declared positive class; span F1 requires tokenization and matching rules. Fix thresholds on validation data and match splits, generators, and evidence. Similar evaluator accuracy can still conceal biased system-level estimates (Godbole and Jia, 2025). Independently audit useful generation after mitigation.

5 Mitigation: Interventions and Assumptions

Figure 4 organizes interventions by generation stage; Table 2 compares targets, access, and resources. Context-aware decoding (CAD) favors supplied evidence, whereas Astute RAG reconciles conflicting knowledge (Shi et al., 2024; Wang et al., 2025). Adhering to a record and answering what is true can require different policies; missing, ignored, and unreliable evidence motivate different interventions.

5.1 Training and alignment

FaithDial supplies curated source-grounded dialogue revisions for generation and critic supervision (Dziri et al., 2022a). Self-Alignment improves self-evaluation through Self-Knowledge Tuning, ranks candidate responses by factuality, and forms response-level preference pairs for direct preference optimization (DPO) (Zhang et al., 2024b). For long answers, atomic-claim scores are averaged before ranking, so training preferences remain response-level.

FactAlign combines response-level Kahneman–Tversky optimization (KTO) with sentence-level fine-grained KTO (fKTO) (Huang and Chen, 2024). Each sentence is treated as a continuation conditioned on the prompt and preceding sentences; binary factuality labels derive from support against retrieved Wikipedia evidence. Unlike DPO, fKTO needs no sentence-level preference pairs. UAlign incorporates confidence and semantic entropy into a reward model before proximal policy optimization (PPO) alignment (Xue et al., 2025). RHIO targets contextual faithfulness: masking retrieval heads produces negative examples, control tokens distinguish faithful from unfaithful continuations, and contrastive decoding subsequently separates them (Huang et al., 2025b). Finer feedback still inherits supervision errors.

5.2 Retrieval and prompting

FLARE retrieves actively when provisional continuations signal uncertainty, while context-faithful prompting changes instructions without retraining (Jiang et al., 2023; Zhou et al., 2023). Retrieval adds information but introduces selection errors and evidence conflicts. Self-RAG trains retrieval and reflection tokens that control when to retrieve and assess relevance, support, and utility; candidate continuations can then be ranked using these signals (Asai et al., 2024). Astute RAG instead uses a training-free, source-aware consolidation procedure: it elicits internal knowledge, compares it with retrieved passages, and iteratively resolves conflicts (Wang et al., 2025). Its reliability judgments remain fallible. These methods require separate audits of retrieval, conflict resolution, and evidence use.

5.3 Decoding and internal intervention

CAD amplifies the difference between context-conditioned and context-free token distributions (Shi et al., 2024). For model parameters θ , next token y_t , context c , query x , prefix $y_{<t}$, and contrast strength $\alpha \geq 0$,

$$p_{\text{CAD}}(y_t) \propto \frac{p_{\theta}(y_t \mid c, x, y_{<t})^{1+\alpha}}{p_{\theta}(y_t \mid x, y_{<t})^{\alpha}}. \quad (1)$$

CAD favors tokens whose probability increases when context is supplied, assuming that context deserves greater weight. DoLa instead subtracts earlier-layer from mature-layer log probabilities within a plausible-token set, dynamically selecting an earlier candidate layer by maximum Jensen–Shannon divergence from the mature layer (Chuang et al., 2024b). It assumes layerwise changes reveal useful parametric knowledge. Both reshape token distributions without independently checking truth.

TruthX learns separate semantic and truth-related latent spaces and edits internal representations along a learned direction (Zhang et al., 2024a). Lookback Lens trains a classifier over context-versus-generation attention ratios and selects candidate continuations with that classifier (Chuang et al., 2024a). Its mitigation closes the detector–generator loop, but also inherits detector errors. Such steering does not establish universal truth representations; architectural transfer needs testing.

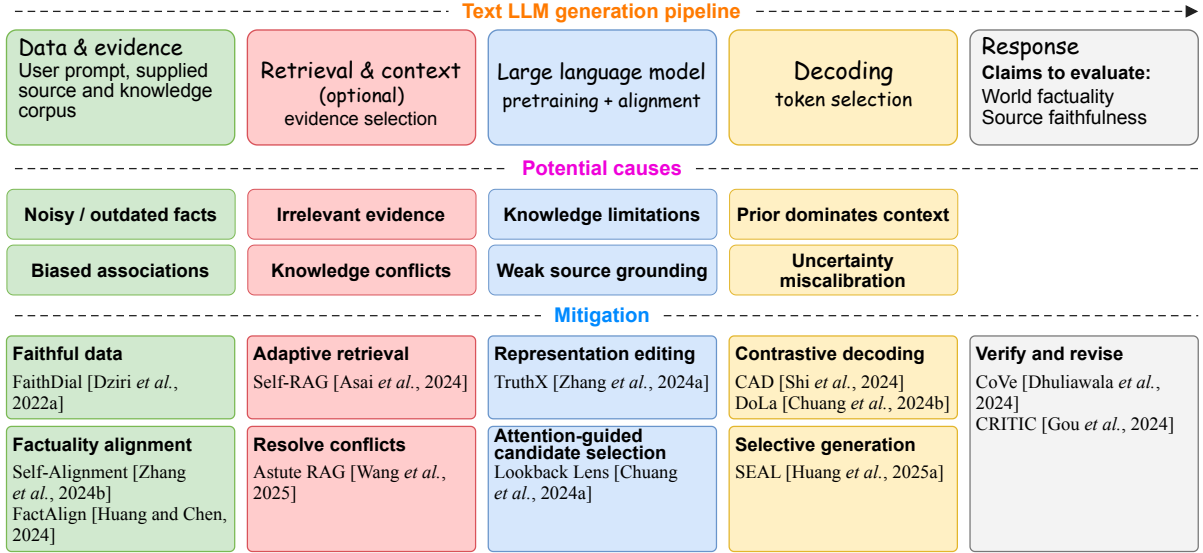


Figure 4: Potential causes and mitigation methods of hallucination in LLMs. Colors group stages of the generation pipeline; hybrid methods may span multiple stages.

Table 2: Representative method trade-offs. Targets describe intended criteria or proxies, not guarantees; resource costs are qualitative.

Family / example	Target	Evidence / model access	Extra resource	Critical boundary
Claim checks: FActScore (Min et al., 2023)	World factuality via source support	Text, corpus, retriever, verifier	Per-claim search and checks	Missing support is not disproof.
Sampling: SelfCheckGPT (Manakul et al., 2023)	World factuality proxy	Repeated outputs; no target internals	Generation and comparisons	Stable falsehoods can agree.
Alignment: FactAlign (Huang and Chen, 2024)	World factuality	Trainable generator; factuality labels	Verification and updates	Verification errors enter training.
RAG: Self-RAG (Asai et al., 2024)	Source support and utility	Trainable model, corpus, reflection labels	Training and retrieval	Reflection is not independent verification.
Decoding: CAD / DoLa (Shi et al., 2024; Chuang et al., 2024b)	Source faithfulness / world factuality	Context-conditioned/free logits / layer outputs	Token-distribution computation	Neither establishes truth.
Revision: CoVe (Dhuliawala et al., 2024)	World factuality	Model calls; no external verifier required	Verification and rewriting	Shared misconceptions can persist.

5.4 Revision and abstention

Chain-of-Verification (CoVe) drafts an answer, generates verification questions, answers them independently, and revises the draft (Dhuliawala et al., 2024). Separating verification from the draft reduces answer contamination, but all stages can still share a parametric misconception. CRITIC adds tool feedback before revision, providing an external correction signal (Gou et al., 2024). WikiChat also grounds dialogue through retrieval and verification, coordinating these operations (Semnani et al., 2023). External feedback remains limited by search and tool reliability.

Abstention can reduce unsupported assertions. SEAL learns a rejection token during training and uses its probability to regularize decoding (Huang et al., 2025a); it is more specific than appending a

refusal instruction. Lower error can reflect fewer answers. Refusal and revision therefore need coverage and information retention alongside precision.

6 Analysis and Future Directions

Reference frames determine whether an intervention succeeds. CAD favors source adherence; Astute RAG reconciles knowledge conflicts, and Stable-RAG addresses passage order (Shi et al., 2024; Wang et al., 2025; Zhang et al., 2026). Following a false source can improve faithfulness without improving world correctness. We propose varying evidence correctness and order with fixed questions and generator, scoring source support and world correctness separately with unknown labels. This tests evidence sensitivity; attribution to retrieval selection additionally requires separate au-

bits of retrieved passages and the generator’s use of them.

Confidence and internal signals are useful proxies, not truth certificates. Semantic entropy targets unstable confabulations (Farquhar et al., 2024), whereas Cheang et al. (2026) find that associated hallucinations resemble factual recall in the representations studied. These findings suggest that uncertainty and probing may detect missing knowledge more readily than confidently recalled misinformation, without establishing a universal detector ranking. We propose testing held-out generators, separating unstable answers from consistent falsehoods and checking whether external verification helps on the latter. Aggregate accuracy cannot identify which failure process a detector recognizes.

Metric improvements must preserve useful information. Response length affects measured precision (Zhao et al., 2025), while ICAT evaluates coverage of expected aspects (Samarinas et al., 2025). On Llama 65B biographies, CoVe’s factor+revise raises FActScore from 55.9 to 71.4 over few-shot prompting, while mean fact count falls from 16.6 to 12.3 (Dhuliawala et al., 2024). Precision gains thus accompany reduced content; this comparison does not isolate a length effect. Report precision, aspect coverage, unique supported claims, and refusals, comparing risk at matched answered-question fractions.

Evaluation must remain independent of optimization. Godbole and Jia (2025) show that similar evaluator accuracy can conceal different class-specific errors and biased system-level estimates. Using the same verifier for optimization and evaluation therefore risks rewarding its preferences. Temporal benchmark errors (Jiang et al., 2026) additionally motivate versioned evidence and label review. Interventions in Figure 4 require independent evaluation against Figure 3’s targets and Table 1’s benchmarks under matched evidence, coverage, and budgets.

Resource trade-offs depend on the intervention stage. FactAlign pays for factuality feedback and parameter updates; Self-RAG combines offline supervision with online retrieval and candidate selection (Huang and Chen, 2024; Asai et al., 2024). CoVe and CRITIC instead add verification and revision calls during use (Dhuliawala et al., 2024; Gou et al., 2024). Training-free operation can still be costly; training expenditure may be amortized

across queries but repeated after domain changes. Report both stages, including tokens, requests, latency, and memory.

Detection and mitigation require joint validation. An intervention can improve generation while making remaining errors harder to detect. Cheng et al. (2025) find that reasoning prompts can reduce hallucinations while obscuring detection cues in their studied settings. For a fixed query distribution, let E denote an erroneous candidate and R the decision to release it. For $P(E) > 0$, the erroneous-release rate satisfies

$$P(E \cap R) = P(E)P(R | E). \quad (2)$$

No independence is assumed. Lower candidate error can be offset by higher false acceptance, so component improvements need not improve the complete system.

We propose crossing generators with fixed detectors to compare baseline, individual, and combined interventions under matched questions, evidence, and budgets. Report erroneous releases per query, error among released answers, and coverage; refusals remain unreleased. At matched coverage, audit corrections, new errors, and retained supported claims (Appendix D).

7 Conclusion

RQ1 requires separating world factuality from source faithfulness and declaring the claim unit and unknown-label policy. For **RQ2**, external evidence enables support checks, whereas sampling and internal signals estimate reliability under distinct access and supervision constraints. For **RQ3**, useful mitigation requires independently verified reliability gains, assessed jointly with informative coverage and computational cost. Future work should test detector–intervention combinations under evidence conflicts and distribution shifts, linking mechanism choice to residual errors.

Limitations

This narrative survey emphasizes English text tasks and ACL-family venues. Venue and keyword filters may omit relevant work; full-text checking was claim-specific rather than uniform. Heterogeneous generators, evidence, judges, and budgets prevent pooled effect estimates. The taxonomy and reporting protocol require prospective validation.

References

- Vaibhav Adlakha, Parishad BehnamGhader, Xing Han Lu, Nicholas Meade, and Siva Reddy. 2024. [Evaluating correctness and faithfulness of instruction-following models for question answering](#). *Transactions of the Association for Computational Linguistics*, 12:681–699.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. [Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection](#). In *The Twelfth International Conference on Learning Representations*.
- Amos Azaria and Tom Mitchell. 2023. [The internal state of an LLM knows when it’s lying](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976, Singapore. Association for Computational Linguistics.
- Yejin Bang, Ziwei Ji, Alan Schelten, Anthony Hartshorn, Tara Fowler, Cheng Zhang, Nicola Cancedda, and Pascale Fung. 2025. [HalluLens: LLM hallucination benchmark](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 24128–24156, Vienna, Austria. Association for Computational Linguistics.
- Forrest Sheng Bao, Miaoran Li, Renyi Qu, Ge Luo, Erana Wan, Yujia Tang, Weisi Fan, Manveer Singh Tamber, Suleman Kazi, Vivek Sourabh, Mike Qi, Ruixuan Tu, Chenyu Xu, Matthew Gonzales, Ofer Mendelevitch, and Amin Ahmad. 2025. [FaithBench: A diverse hallucination benchmark for summarization by Modern LLMs](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pages 448–461, Albuquerque, New Mexico. Association for Computational Linguistics.
- Baolong Bi, Shaohan Huang, Yiwei Wang, Tianchi Yang, Zihan Zhang, Haizhen Huang, Lingrui Mei, Junfeng Fang, Zehao Li, Furu Wei, Weiwei Deng, Feng Sun, Qi Zhang, and Shenghua Liu. 2025. [Context-DPO: Aligning language models for context-faithfulness](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 10280–10300, Vienna, Austria. Association for Computational Linguistics.
- Meng Cao, Yue Dong, and Jackie Cheung. 2022. [Hallucinated but factual! inspecting the factuality of hallucinations in abstractive summarization](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3340–3354, Dublin, Ireland. Association for Computational Linguistics.
- Arie Cattan, Paul Roit, Shiyue Zhang, David Wan, Roei Aharoni, Idan Szpektor, Mohit Bansal, and Ido Dagan. 2026. [Localizing factual inconsistencies in attributable text generation](#). *Transactions of the Association for Computational Linguistics*, 14:100–123.
- Chi Seng Cheang, Hou Pong Chan, Wenxuan Zhang, and Yang Deng. 2026. [Do LLMs really know what they don’t know? internal states mainly reflect knowledge recall rather than truthfulness](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 713–730, San Diego, California, United States. Association for Computational Linguistics.
- Jiawei Chen, Hongyu Lin, Xianpei Han, and Le Sun. 2024. [Benchmarking Large Language Models in Retrieval-Augmented Generation](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 17754–17762.
- Wenbo Chen, Veena Padmanabhan, Tootiya Giyahchi, Elaine Wong, and Leman Akoglu. 2026. [Rethinking evaluation for LLM hallucination detection: A desiderata, a new RAG-based benchmark, new insights](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14912–14931, San Diego, California, United States. Association for Computational Linguistics.
- Jiahao Cheng, Tiancheng Su, Jia Yuan, Guoxiu He, Jiawei Liu, Xinqi Tao, Jingwen Xie, and Huaxia Li. 2025. [Chain-of-thought prompting obscures hallucination cues in large language models: An empirical evaluation](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 1272–1305, Suzhou, China. Association for Computational Linguistics.
- Yung-Sung Chuang, Linlu Qiu, Cheng-Yu Hsieh, Ranjay Krishna, Yoon Kim, and James R. Glass. 2024a. [Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 1419–1436, Miami, Florida, USA. Association for Computational Linguistics.
- Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, and Pengcheng He. 2024b. [DoLa: Decoding by Contrasting Layers Improves Factuality in Large Language Models](#). In *The Twelfth International Conference on Learning Representations*.
- Roi Cohen, May Hamri, Mor Geva, and Amir Globerson. 2023. [LM vs LM: Detecting factual errors via cross examination](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12621–12640, Singapore. Association for Computational Linguistics.
- Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2024. [Chain-of-verification reduces hallucination in large language models](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3563–3578, Bangkok, Thailand. Association for Computational Linguistics.
- Nouha Dziri, Ehsan Kamalloo, Sivan Milton, Omar Zaiane, Mo Yu, Edoardo M. Ponti, and Siva

- Reddy. 2022a. [FaithDial: A faithful benchmark for information-seeking dialogue](#). *Transactions of the Association for Computational Linguistics*, 10:1473–1490.
- Nouha Dziri, Hannah Rashkin, Tal Linzen, and David Reitter. 2022b. [Evaluating attribution in dialogue systems: The BEGIN benchmark](#). *Transactions of the Association for Computational Linguistics*, 10:1066–1083.
- Alexander Fabbri, Chien-Sheng Wu, Wenhao Liu, and Caiming Xiong. 2022. [QAFactEval: Improved QA-based factual consistency evaluation for summarization](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2587–2601, Seattle, United States. Association for Computational Linguistics.
- Cunhang Fan, Jun Zhang, Xue Zhang, Shuai Zhang, Zhao Lv, Jianhua Tao, and Zhengqi Wen. 2026. [ReFL: Reflective feedback learning for hallucination detection of large language models](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 19648–19665, San Diego, California, United States. Association for Computational Linguistics.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. 2024. [Detecting hallucinations in large language models using semantic entropy](#). *Nature*, 630(8017):625–630.
- Farima Fatahi Bayat, Lechen Zhang, Sheza Munir, and Lu Wang. 2025. [FactBench: A dynamic benchmark for in-the-wild language model factuality evaluation](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 33090–33110, Vienna, Austria. Association for Computational Linguistics.
- Shangbin Feng, Vidhisha Balachandran, Yuyang Bai, and Yulia Tsvetkov. 2023. [FactKB: Generalizable factuality evaluation using language models enhanced with factual knowledge](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 933–952, Singapore. Association for Computational Linguistics.
- Shangbin Feng, Weijia Shi, Yike Wang, Wenxuan Ding, Vidhisha Balachandran, and Yulia Tsvetkov. 2024. [Don’t hallucinate, abstain: Identifying LLM knowledge gaps via multi-LLM collaboration](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14664–14690, Bangkok, Thailand. Association for Computational Linguistics.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. [Enabling large language models to generate text with citations](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488, Singapore. Association for Computational Linguistics.
- Zorik Gekhman, Jonathan Herzig, Roei Aharoni, Chen Elkind, and Idan Szpektor. 2023. [TrueTeacher: Learning factual consistency evaluation with large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2053–2070, Singapore. Association for Computational Linguistics.
- Zorik Gekhman, Gal Yona, Roei Aharoni, Matan Eyal, Amir Feder, Roi Reichart, and Jonathan Herzig. 2024. [Does fine-tuning LLMs on new knowledge encourage hallucinations?](#) In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 7765–7784, Miami, Florida, USA. Association for Computational Linguistics.
- Aryo Pradipta Gema, Chen Jin, Ahmed Abdulaal, Tom Diethe, Philip Alexander Teare, Beatrice Alex, Pasquale Minervini, and Amrutha Saseendran. 2025. [DeCoRe: Decoding by contrasting retrieval heads to mitigate hallucinations](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 10003–10039, Suzhou, China. Association for Computational Linguistics.
- Ameya Godbole and Robin Jia. 2025. [Verify with caution: The pitfalls of relying on imperfect factuality metrics](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 22889–22912, Vienna, Austria. Association for Computational Linguistics.
- Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. 2024. [CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing](#). In *The Twelfth International Conference on Learning Representations*.
- Sapir Harary, Eran Hirsch, Aviv Slobodkin, David Wan, Mohit Bansal, and Ido Dagan. 2026. [PrefixNLI: Detecting factual inconsistencies as soon as they arise](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1414–1433, San Diego, California, United States. Association for Computational Linguistics.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. 2022. [TRUE: Re-evaluating factual consistency evaluation](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3905–3920, Seattle, United States. Association for Computational Linguistics.
- Chao-Wei Huang and Yun-Nung Chen. 2024. [FactAlign: Long-form factuality alignment of large language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 16363–16375, Miami, Florida, USA. Association for Computational Linguistics.

- Lei Huang, Xiaocheng Feng, Weitao Ma, Yuchun Fan, Xiachong Feng, Yuxuan Gu, Yangfan Ye, Liang Zhao, Weihong Zhong, Baoxin Wang, Dayong Wu, Guoping Hu, Lingpeng Kong, Tong Xiao, Ting Liu, and Bing Qin. 2025a. [Alleviating hallucinations from knowledge misalignment in large language models via selective abstention learning](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 24564–24579, Vienna, Austria. Association for Computational Linguistics.
- Lei Huang, Xiaocheng Feng, Weitao Ma, Yuchun Fan, Xiachong Feng, Yangfan Ye, Weihong Zhong, Yuxuan Gu, Baoxin Wang, Dayong Wu, Guoping Hu, and Bing Qin. 2025b. [Improving contextual faithfulness of large language models via retrieval heads-induced optimization](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16896–16913, Vienna, Austria. Association for Computational Linguistics.
- Denis Janiak, Jakub Binkowski, Albert Sawczyn, Bogdan Gabrys, Ravid Shwartz-Ziv, and Tomasz Jan Kajdanowicz. 2025. [The illusion of progress: Re-evaluating hallucination detection in LLMs](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 34728–34745, Suzhou, China. Association for Computational Linguistics.
- Ziwei Ji, Yuzhe Gu, Wenwei Zhang, Chengqi Lyu, Dahua Lin, and Kai Chen. 2024. [ANAH: Analytical annotation of hallucinations in large language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8135–8158, Bangkok, Thailand. Association for Computational Linguistics.
- Che Jiang, Biqing Qi, Xiangyu Hong, Dayuan Fu, Yang Cheng, Fandong Meng, Mo Yu, Bowen Zhou, and Jie Zhou. 2024. [On large language models’ hallucination with regard to known facts](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1041–1053, Mexico City, Mexico. Association for Computational Linguistics.
- Xunyi Jiang, Dingyi Chang, Julian McAuley, and Xin Xu. 2026. [When benchmarks age: Temporal misalignment through large language model factuality evaluation](#). In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 500–512, Rabat, Morocco. Association for Computational Linguistics.
- Zhengbao Jiang, Frank Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. [Active retrieval augmented generation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7969–7992, Singapore. Association for Computational Linguistics.
- Zhengping Jiang, Jingyu Zhang, Nathaniel Weir, Seth Ebner, Miriam Wanner, Kate Sanders, Daniel Khashabi, Anqi Liu, and Benjamin Van Durme. 2025. [Core: Robust factual precision with informative subclaim identification](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 19833–19856, Vienna, Austria. Association for Computational Linguistics.
- Katie Kang, Eric Wallace, Claire Tomlin, Aviral Kumar, and Sergey Levine. 2025. [Unfamiliar finetuning examples control how language models hallucinate](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3600–3612, Albuquerque, New Mexico. Association for Computational Linguistics.
- Anant Khandelwal, Manish Gupta, and Puneet Agrawal. 2025. [CoCoA: Confidence- and context-aware adaptive decoding for resolving knowledge conflicts in large language models](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 6835–6855, Suzhou, China. Association for Computational Linguistics.
- Kalpesh Krishna, Erin Bransom, Bailey Kuehl, Mohit Iyyer, Pradeep Dasigi, Arman Cohan, and Kyle Lo. 2023. [LongEval: Guidelines for human evaluation of faithfulness in long-form summarization](#). In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1650–1669, Dubrovnik, Croatia. Association for Computational Linguistics.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. [Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation](#). In *The Eleventh International Conference on Learning Representations*.
- Joosung Lee, Cheonbok Park, Hwiyeol Jo, Jeonghoon Kim, Joonsuk Park, and Kang Min Yoo. 2026. [Enhancing hallucination detection via future context](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 731–749, San Diego, California, United States. Association for Computational Linguistics.
- Junyi Li, Jie Chen, Ruiyang Ren, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. 2024. [The dawn after the dark: An empirical study on factuality hallucination in large language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10879–10899, Bangkok, Thailand. Association for Computational Linguistics.
- Junyi Li, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. 2023. [HaluEval: A large-scale hallucination evaluation benchmark for large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6449–6464, Singapore. Association for Computational Linguistics.

- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. [TruthfulQA: Measuring how models mimic human falsehoods](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.
- Hanchao Liu, Wenyuan Xue, Yifei Chen, Dapeng Chen, Xiutian Zhao, Ke Wang, Liping Hou, Rongjun Li, and Wei Peng. 2024a. [A survey on hallucination in large vision-language models](#). *arXiv preprint arXiv:2402.00253*. Design reference; supplied version v2.
- Hanmeng Liu, Zhizhang Fu, Mengru Ding, Ruoxi Ning, Chaoli Zhang, Xiaozhang Liu, and Yue Zhang. 2025. [Logical reasoning in large language models: A survey](#). *arXiv preprint arXiv:2502.09100*. Design reference; supplied version v1.
- Junteng Liu, Shiqi Chen, Yu Cheng, and Junxian He. 2024b. [On the universal truthfulness hyperplane inside LLMs](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 18199–18224, Miami, Florida, USA. Association for Computational Linguistics.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024c. [Lost in the middle: How language models use long contexts](#). *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Liang Ma, Shuyang Cao, Robert L Logan IV, Di Lu, Shihao Ran, Ke Zhang, Joel Tetreault, and Alejandro Jaimes. 2023. [BUMP: A benchmark of unfaithful minimal pairs for meta-evaluation of faithfulness metrics](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12788–12812, Toronto, Canada. Association for Computational Linguistics.
- Matéo Mahaut, Laura Aina, Paula Czarnowska, Momchil Hardalov, Thomas Müller, and Lluís Marquez. 2024. [Factual confidence of LLMs: on reliability and robustness of current estimators](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4554–4570, Bangkok, Thailand. Association for Computational Linguistics.
- Chaitanya Malaviya, Subin Lee, Sihao Chen, Elizabeth Sieber, Mark Yatskar, and Dan Roth. 2024. [ExpertQA: Expert-curated questions and attributed answers](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3025–3045, Mexico City, Mexico. Association for Computational Linguistics.
- Potsawee Manakul, Adian Liusie, and Mark Gales. 2023. [SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9004–9017, Singapore. Association for Computational Linguistics.
- Radu Marinescu, Debarun Bhattacharjya, Junkyu Lee, Tigran T. Tchirakian, Javier Carnerero-Cano, Yufang Hou, Elizabeth M. Daly, and Alessandra Pascale. 2025. [FactReasoner: A probabilistic approach to long-form factuality assessment for large language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 14547–14577, Suzhou, China. Association for Computational Linguistics.
- Sewon Min, Kalpesh Krishna, Xinxin Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. [FactScore: Fine-grained atomic evaluation of factual precision in long form text generation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore. Association for Computational Linguistics.
- Dor Muhlgay, Ori Ram, Inbal Magar, Yoav Levine, Nir Ratner, Yonatan Belinkov, Omri Abend, Kevin Leyton-Brown, Amnon Shashua, and Yoav Shoham. 2024. [Generating benchmarks for factuality evaluation of language models](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 49–66, St. Julian’s, Malta. Association for Computational Linguistics.
- Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. [RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878, Bangkok, Thailand. Association for Computational Linguistics.
- Abhilasha Ravichander, Shruti Ghela, David Wadden, and Yejin Choi. 2025. [HALoGEN: Fantastic LLM hallucinations and where to find them](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1402–1425, Vienna, Austria. Association for Computational Linguistics.
- Chris Samarin, Alexander Krubner, Alireza Salemi, Youngwoo Kim, and Hamed Zamani. 2025. [Beyond factual accuracy: Evaluating coverage of diverse factual information in long-form text generation](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 13468–13482, Vienna, Austria. Association for Computational Linguistics.
- Jakob Schuster, Vagrant Gautam, and Katja Markert. 2026. [Whose facts win? LLM source preferences under knowledge conflicts](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 29430–29459, San Diego, California, United States. Association for Computational Linguistics.

- Alessandro Scirè, Karim Ghonim, and Roberto Navigli. 2024. [FENICE: Factuality evaluation of summarization based on natural language inference and claim extraction](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 14148–14161, Bangkok, Thailand. Association for Computational Linguistics.
- Sina Semnani, Violet Yao, Heidi Zhang, and Monica Lam. 2023. [WikiChat: Stopping the hallucination of large language model chatbots by few-shot grounding on Wikipedia](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 2387–2413, Singapore. Association for Computational Linguistics.
- Giovanni Servadio, Alessandro De Bellis, Dario Di Palma, Vito Walter Anelli, and Tommaso Di Noia. 2025. [Are the hidden states hiding something? testing the limits of factuality-encoding capabilities in LLMs](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6089–6104, Vienna, Austria. Association for Computational Linguistics.
- Weijia Shi, Xiaochuang Han, Mike Lewis, Yulia Tsvetkov, Luke Zettlemoyer, and Wen-tau Yih. 2024. [Trusting your evidence: Hallucinate less with context-aware decoding](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pages 783–791, Mexico City, Mexico. Association for Computational Linguistics.
- Shuzheng Si, Qingyi Wang, Haozhe Zhao, Yuzhuo Bai, Guanqiao Chen, Kangyang Luo, Gang Chen, Fan-chao Qi, Minjia Zhang, Baobao Chang, and Maosong Sun. 2026. [FaithLens: Detecting and explaining faithfulness hallucination](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 14068–14099, San Diego, California, United States. Association for Computational Linguistics.
- Linxin Song, Taiwei Shi, and Jieyu Zhao. 2025. [The hallucination tax of reinforcement finetuning](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 2105–2120, Suzhou, China. Association for Computational Linguistics.
- Yixiao Song, Yekyung Kim, and Mohit Iyyer. 2024. [VeriScore: Evaluating the factuality of verifiable claims in long-form text generation](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 9447–9474, Miami, Florida, USA. Association for Computational Linguistics.
- Liyan Tang, Tanya Goyal, Alex Fabbri, Philippe Laban, Jiacheng Xu, Semih Yavuz, Wojciech Kryscinski, Justin Rousseau, and Greg Durrett. 2023. [Understanding factual errors in summarization: Errors, summarizers, datasets, error detectors](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11626–11644, Toronto, Canada. Association for Computational Linguistics.
- Liyan Tang, Igor Shalymov, Amy Wong, Jon Burnsky, Jake Vincent, Yu’an Yang, Siffi Singh, Song Feng, Hwanjun Song, Hang Su, Lijia Sun, Yi Zhang, Saab Mansour, and Kathleen McKeown. 2024. [TofuEval: Evaluating hallucinations of LLMs on topic-focused dialogue summarization](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4455–4480, Mexico City, Mexico. Association for Computational Linguistics.
- Fei Wang, Xingchen Wan, Ruoxi Sun, Jiefeng Chen, and Sercan O Arik. 2025. [Astute RAG: Overcoming imperfect retrieval augmentation and knowledge conflicts for large language models](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 30553–30571, Vienna, Austria. Association for Computational Linguistics.
- Yuxia Wang, Minghan Wang, Muhammad Arslan Manzoor, Fei Liu, Georgi Nenkov Georgiev, Rocktim Jyoti Das, and Preslav Nakov. 2024. [Factuality of large language models: A survey](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 19519–19529, Miami, Florida, USA. Association for Computational Linguistics.
- Di Wu, Jia-Chen Gu, Fan Yin, Nanyun Peng, and Kai-Wei Chang. 2024. [Synchronous faithfulness monitoring for trustworthy retrieval-augmented generation](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 9390–9406, Miami, Florida, USA. Association for Computational Linguistics.
- Wenhao Wu, Wei Li, Xinyan Xiao, Jiachen Liu, Sujian Li, and Yajuan Lyu. 2023. [WeCheck: Strong factual consistency checker via weakly supervised learning](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 307–321, Toronto, Canada. Association for Computational Linguistics.
- Boyang Xue, Fei Mi, Qi Zhu, Hongru Wang, Rui Wang, Sheng Wang, Erxin Yu, Xuming Hu, and Kam-Fai Wong. 2025. [UAlign: Leveraging uncertainty estimations for factuality alignment on large language models](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6002–6024, Vienna, Austria. Association for Computational Linguistics.
- Xiaodong Yu, Hao Cheng, Xiaodong Liu, Dan Roth, and Jianfeng Gao. 2024. [ReEval: Automatic hallucination evaluation for retrieval-augmented large language models via transferable adversarial attacks](#). In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 1333–1351, Mexico City, Mexico. Association for Computational Linguistics.
- Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. [AlignScore: Evaluating factual consistency](#)

- with a unified alignment function. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11328–11348, Toronto, Canada. Association for Computational Linguistics.
- Jiaxin Zhang, Zhuohang Li, Kamalika Das, Bradley Malin, and Sricharan Kumar. 2023a. [SAC³: Reliable hallucination detection in black-box language models via semantic-aware cross-check consistency](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 15445–15458, Singapore. Association for Computational Linguistics.
- Qianchi Zhang, Hainan Zhang, Liang Pang, Hong-Wei Zheng, and Zhiming Zheng. 2026. [Stable-RAG: Mitigating retrieval-permutation-induced hallucinations in retrieval-augmented generation](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 25907–25926, San Diego, California, United States. Association for Computational Linguistics.
- Qinggang Zhang, Zhishang Xiang, Yilin Xiao, Le Wang, Junhui Li, Xinrun Wang, and Jinsong Su. 2025. [FaithfulRAG: Fact-level conflict modeling for context-faithful retrieval-augmented generation](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 21863–21882, Vienna, Austria. Association for Computational Linguistics.
- Shaolei Zhang, Tian Yu, and Yang Feng. 2024a. [TruthX: Alleviating hallucinations by editing large language models in truthful space](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8908–8949, Bangkok, Thailand. Association for Computational Linguistics.
- Tianhang Zhang, Lin Qiu, Qipeng Guo, Cheng Deng, Yue Zhang, Zheng Zhang, Chenghu Zhou, Xinbing Wang, and Luoyi Fu. 2023b. [Enhancing uncertainty-based hallucination detection with stronger focus](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 915–932, Singapore. Association for Computational Linguistics.
- Xiaoying Zhang, Baolin Peng, Ye Tian, Jingyan Zhou, Lifeng Jin, Linfeng Song, Haitao Mi, and Helen Meng. 2024b. [Self-alignment for factuality: Mitigating hallucinations in LLMs via self-evaluation](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1946–1965, Bangkok, Thailand. Association for Computational Linguistics.
- James Xu Zhao, Jimmy Z.J. Liu, Bryan Hooi, and See-Kiong Ng. 2025. [How does response length affect long-form factuality](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 3102–3125, Vienna, Austria. Association for Computational Linguistics.
- Wenxuan Zhou, Sheng Zhang, Hoifung Poon, and Muhao Chen. 2023. [Context-faithful prompting for large language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 14544–14556, Singapore. Association for Computational Linguistics.

A Selection, Provenance, and Reading Depth

This is a targeted narrative survey with a metadata snapshot of 15 September 2026, not an exhaustive systematic review or a meta-analysis. Discovery used official ACL Anthology collections for ACL, EMNLP, NAACL, EACL, Findings, and TACL across 2022–2026. Twenty-five collections were available; five requested year/venue combinations returned unavailable responses and were logged as such. A broad title filter for hallucination, factuality, faithfulness, attribution, uncertainty, retrieval, and related terms returned 1,766 records. Three targeted Anthology additions and seven external mechanism-defining studies produced 1,776 candidate records.

Selection prioritized direct relevance to text factuality or source faithfulness, distinct mechanisms, benchmark diversity, and evidence that qualified common assumptions. The working library contains 87 studies: 6 from 2022, 20 from 2023, 29 from 2024, 22 from 2025, and 10 from 2026. Eighty have Anthology records; the seven external records were checked against primary publication metadata. Canonical published versions were preferred over duplicate preprints. DOI and normalized-title checks found no exact duplicates among selected records; this is not a guarantee about near-duplicates in the reserve set. Automatic title triage excluded 401 candidates, and 1,288 remain in reserve without individual abstract assessment. These counts describe the actual workflow, not full-text exclusion decisions.

All 87 selected titles and abstracts were screened. Initial preparation inspected selected passages of 15 downloaded representative papers; drafting added targeted primary-source checks for the detailed detection, evaluation, and mitigation claims. This does not mean every selected paper was read in full. The companion evidence ledgers record claim-specific sections, access assumptions, and interpretive limits. The broad supplementary catalogue below distinguishes mechanisms without reproducing performance rankings. No experiments, independent benchmark replications, or statistical pooling were performed.

The synthesis cites 86 research studies from the 87-study discovery library, including references within the figures. Two additional surveys provide the visual-design sources: the layouts of Figures 1, 3, and 4 and Table 1 follow Liu et al. (2024a); the

structure tree in Figure 2 follows Liu et al. (2025). Figure 1 uses synthetic interactions to illustrate source-support distinctions. Figure 4 summarizes proposed mechanisms across studies rather than an experimentally identified causal decomposition.

Coding rules. For each study, record the correctness target, scoring unit, evidence source, model signals, supervision, and intervention stage. Assign detection branches by decision evidence and mitigation branches by the operation that changes generation; separate training from inference requirements. For example, Lookback Lens is coded as attention-based detection and candidate-selection mitigation. RHIO is coded under training for induced negatives and under decoding for its contrastive intervention. Record source-faithfulness targets separately from world-factuality targets, and distinguish target-generator access from proxy or verifier access. These rules make classification choices explicit without forcing hybrid methods into exclusive categories.

B Benchmark Units and Protocol Caveats

Table 3 supplies the counting and interpretation details behind Table 1. A benchmark may support both a description of generated errors and evaluation of a detector, but those uses require different denominators and sampling assumptions. In particular, reporting error frequencies within deliberately selected difficult examples does not estimate a generator’s error prevalence on ordinary requests.

C Supplementary Comparative Evidence

Tables 4–7 extend the main synthesis with complementary studies. The final column states the inference boundary used in this survey; it is not an assertion that the original authors omitted every proposed test. These studies expand coverage of mechanisms and failure conditions rather than form a ranked list. Descriptive entries are grounded in the recorded primary metadata and selected source passages, with reading depth retained in the repository.

Table 3: Evaluation units, protocols, and limitations of the benchmarks in Table 1.

Benchmark	Unit, protocol, and limitation
TruthfulQA	The original 2022 benchmark contains 817 questions. Generation measures truthfulness/informativeness; MC1 measures single-true accuracy, and MC2 normalized probability mass assigned to true answers. The official January 2025 update introduced two-option choices and revised the question set. Report the dataset version and MC protocol separately; answer selection is not detector classification.
HaluEval	The 35,000 examples combine task-specific generated examples with a human-annotated general-query portion. Artificial and naturally generated errors should be reported separately rather than treated as one deployment distribution.
ANAH	Approximately 12,000 annotated sentences provide error types, evidence, and corrections. Generative versus discriminative annotators describe detector implementations; their task remains prediction of annotation labels.
RAGTruth	The 17,790 responses come from 2,965 prompts and six generators, not 17,790 independent documents. Response classification and span localization have different matching rules and must be evaluated separately.
ExpertQA	The 2,177 expert-curated questions support factuality and attribution analysis. Correctness and cited-source support are separate judgments; exact evaluation subsets and answer-generation conditions matter.
TofuEval	Section 3.3 reports 1,500 initial summaries, removal of 23, and 1,479 retained summaries. Those values are arithmetically inconsistent. We retain the reported 1,479 rather than invent a corrected dataset size. Topic relevance and completeness are separate from factual consistency.
HALoGEN	The 10,923 prompts cover all nine domains, including programming. That is the complete-suite count, not a verified text-only subset size. Different domain verifiers do not constitute one universal human truth oracle.
FaithBench	The 750 summaries derive from 75 passages and ten models, selected using detector disagreement. This supports difficult-case detector analysis; error frequencies are conditional on selection, not population prevalence.
FactBench	The 1,000-prompt snapshot uses user-derived prompts and fine-grained verification. Supported, unsupported, and undecidable units require explicit aggregation rules. A dynamic benchmark needs a fixed version for reproducible comparisons.

Table 4: Additional benchmarks and evaluation protocols for hallucination in LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
TRUE (Honovich et al., 2022)	Standardizes factual-consistency datasets across tasks and evaluates metrics at the example level.	Instance discrimination differs from system-level correlation. Combined existing tasks do not estimate real-world hallucination prevalence.
AggreFact analysis (Tang et al., 2023)	Stratifies summarization error annotations by generating model and error type; metric gains vary across strata.	Pooled scores can hide weaknesses on newer generators or rare error types.
BUMP (Ma et al., 2023)	Uses minimally differing human-written summary pairs with one introduced error.	A controlled error-sensitivity probe complements natural generation data; minimal pairs do not reproduce the full distribution of model errors.
ALCE (Gao et al., 2023)	Evaluates answer quality and citation quality in end-to-end retrieval and cited generation.	Citation presence, citation support, and factual correctness are distinct criteria. A plausible citation is not by itself evidence.
LongEval (Krishna et al., 2023)	Studies fine-grained human judgments and partial annotation for long summaries.	Fine-grained annotation requires agreement and coverage reporting; partial annotation does not exhaustively verify all claims.
FACTOR (Muhlgay et al., 2024)	Transforms factual corpora into true versus false continuation choices.	Controlled factual choice evaluates a different task from open-ended generation and error localization.
ReEval (Yu et al., 2024)	Perturbs evidence to create dynamic tests of evidence use under retrieval.	Static accuracy may reflect memorized knowledge rather than source adherence; counterfactual evidence requires an explicit faithfulness target.
Core (Jiang et al., 2025)	Filters redundant or uninformative subclaims before factual precision aggregation.	Trivial repetitions can inflate precision; informativeness and coverage are complementary criteria.
TRIVIA+ (Chen et al., 2026)	Adds long-context RAG detection data and controlled sample-dependent label noise.	Long-context and noisy-label tests complement evaluation on clean labels.
RGB (Chen et al., 2024)	Separates noise robustness, negative rejection, information integration, and counterfactual robustness in English/Chinese RAG.	Retrieval conditions define distinct RAG abilities, not one interchangeable hallucination score.

Table 5: Additional hallucination detection and explanation methods for LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
WeCheck (Wu et al., 2023)	Aggregates weak labels over actual generated samples, then learns a noise-aware classifier.	Supervision quality and error distribution distinguish a learned consistency model from zero-shot entailment.
FactKB (Feng et al., 2023)	Uses knowledge-base factual pretraining with entity/relation objectives for factuality evaluation.	Structured supervision targets relation errors; broader transfer requires separate evidence.
TrueTeacher (Gekhman et al., 2023)	Uses LLM-labeled model-generated summaries to train a smaller consistency evaluator.	Distillation trades annotation cost for teacher-label dependence; generated errors differ from artificial perturbations.
FENICE (Scirè et al., 2024)	Extracts atomic claims and aligns them to a source using natural language inference (NLI).	Local alignment improves interpretability, while claim extraction and entailment both remain possible error sources.
FactReasoner (Marinescu et al., 2025)	Models evidence–claim logical relationships probabilistically after decomposition/retrieval.	Joint reasoning differs from independently verifying claims; its result still depends on evidence retrieval and relation estimates.
Future context (Lee et al., 2026)	Samples continuations and uses them as clues for black-box hallucination detection.	Generated continuations extend consistency checking without supplying independent external verification.
FaithLens (Si et al., 2026)	Trains detection and explanations using filtered synthetic data, supervised fine-tuning (SFT), and rule-based reinforcement learning.	Explanation quality and decision accuracy require separate validation; fluency alone does not establish explanation faithfulness.
QASemConsistency (Cattan et al., 2026)	Decomposes claims into predicate–argument QA pairs for fine-grained support localization.	Semantic error localization supplies a finer target than response-level binary classification.

Table 6: Studies of hallucination mechanisms and reliability in LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
Factual confidence (Mahaut et al., 2024)	Compares confidence estimators and checks stability under meaning-preserving input changes.	Confidence can be unstable across paraphrases; access and supervision requirements differ.
New-knowledge fine-tuning (Gekhman et al., 2024)	Controlled closed-book QA experiments vary the amount of unfamiliar factual information in SFT.	The findings motivate knowledge-aware supervision within the studied fine-tuning setup.
Known facts (Jiang et al., 2024)	Examines layer-wise prediction dynamics when questions about the same fact yield correct versus incorrect answers.	Errors can reflect knowledge access rather than absence; measured patterns do not prove universal causal mechanisms.
Lost in the Middle (Liu et al., 2024c)	Evaluates how position of relevant evidence affects long-context use.	Evidence being present is insufficient for reliable use; its tasks are not themselves a comprehensive hallucination taxonomy.
Hidden-state limits (Servadio et al., 2025)	Tests factuality encoding on more realistic, model-dependent generated data.	Accuracy on synthetic true/false facts does not establish probe generalization to target-model outputs.
Unfamiliar supervision (Kang et al., 2025)	Relates hallucination response patterns to supervision of unfamiliar fine-tuning examples, including reward-model errors.	Factuality alignment depends on how unknowns and reward errors are labeled.
Illusion of Progress (Janiak et al., 2025)	Examines lexical-overlap labels and response-length heuristics in detector evaluation.	Semantic validation and simple baselines remain necessary; an LLM judge is not automatically a gold standard.
CoT obscures cues (Cheng et al., 2025)	Measures how reasoning prompts shift hidden/probability signals and detector effectiveness.	A generation intervention shifts the detector’s input distribution, making calibration transfer uncertain.
Hallucination tax (Song et al., 2025)	Studies refusal degradation after reinforcement fine-tuning, primarily on unanswerable mathematics, with reported transfer to factual QA.	Evidence is not a universal result for every factual-generation task.
Source preferences (Schuster et al., 2026)	Controlled synthetic-source study of source preferences and repetition in conflicting English evidence.	Frequency and source cues influence conflict resolution; synthetic-source preference is not proof of real-world source credibility.

Table 7: Additional hallucination mitigation methods for LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
FaithfulRAG (Zhang et al., 2025)	Models fact-level conflicts between retrieved and parametric knowledge before response generation.	Explicit reconciliation complements source preference; source adherence and world correctness remain distinct.
CoCoA (Khandelwal et al., 2025)	Adapts token-level decoding using confidence/context distribution differences.	Adaptive and fixed contrast differ in conflict sensitivity; low-conflict behavior also matters.
Context-DPO (Bi et al., 2025)	Constructs preferences from faithful versus stubborn responses under knowledge conflict.	A context-faithfulness objective differs from truthfulness with respect to external facts.

D A Common Reporting Protocol

The following protocol is a synthesis of the comparisons in Sections 3–6, not a benchmark run. It can be used to evaluate whether an intervention in Figure 4 solves the kind of error illustrated in Figure 1 under the target in Figure 3.

D.1 Fix the target before selecting a metric

Specify whether the target is world correctness, source support, or both; the answer unit; and the treatment of unknown or unverifiable claims. Record the generator/version, prompts, retrieval corpus/date, evidence truncation, decoding parameters, and judge/version. Use the same evidence and question distribution across systems where possible. If a method changes retrieval, separate retrieval quality from generation conditional on the retrieved passages.

For a nonempty extracted claim set C and fixed evidence E , a schematic support precision is

$$P_{\text{support}}(C, E) = \frac{\sum_{c \in C} \mathbf{1}[E \models c]}{|C|}. \quad (3)$$

This expression describes an aggregation principle, not the complete implementation of every factuality metric. Extraction determines the denominator; duplicate, trivial, subjective, and undecidable claims need explicit handling. An empty answer is not assigned perfect useful factuality by this definition. FActScore and VeriScore motivate claim-level analysis, while Core highlights the role of informative subclaims (Min et al., 2023; Song et al., 2024; Jiang et al., 2025).

D.2 Separate detection, localization, and generation quality

For binary detection, report class prevalence, confusion counts, precision/recall/F1, and balanced accuracy, $\frac{1}{2}(\text{TPR} + \text{TNR})$, where appropriate. Fix thresholds on independent validation data. For

spans, specify tokenization and exact versus overlap matching. For continuous scores, report discrimination and calibration separately; neither establishes the truth of an individual answer. Inspect false-positive and false-negative examples across error types, evidence lengths, and generator families.

For generation, pair factuality with unique supported content and required-aspect coverage. A source-faithful answer may still omit the answer to the question. Human auditing should sample both high- and low-scoring outputs, use explicit instructions, and report disagreement or adjudication. LongEval provides relevant human-evaluation guidance (Krishna et al., 2023). Avoid pooling scalar scores whose claim extraction, evidence, or unknown-label policy differs.

D.3 Measure selection and revision effects

Let A_τ be the answers retained by threshold τ among N questions, and let e_i indicate an erroneous answer under the declared reference frame. For nonempty A_τ , define

$$\begin{aligned} \text{Coverage}(\tau) &= \frac{|A_\tau|}{N}, \\ \text{Risk}(\tau) &= \frac{\sum_{i \in A_\tau} e_i}{|A_\tau|}. \end{aligned} \quad (4)$$

Comparing risk at matched coverage prevents a system from appearing universally better merely by refusing more questions. This response-level selective-evaluation definition is distinct from SEAL’s token-level training objective. For revision, count corrected errors, newly introduced errors, retained supported claims, and required information lost. Measure both before and after revision on the same initial drafts.

D.4 Audit independence, robustness, and cost

A verifier used for training or candidate selection should not be the sole final judge. Use indepen-

Table 8: Access requirements and computational resources of representative methods. Resource descriptions are qualitative and do not imply measured cost rankings.

Family / example	Required access or supervision	Principal extra resource	Important boundary
Source alignment / AlignScore	Source and output text; trained auxiliary model	Verifier inference over chunks	Auxiliary-model errors and evidence truncation
Claim verification / FActScore	Claim extraction, knowledge source, retrieval and verifier	Search and per-claim checks	Retrieval failure differs from disproof
Self-checking / Self-CheckGPT	Repeated outputs; no generator internals	Additional generation and comparisons	Repeated falsehood can appear consistent
Semantic entropy	Samples and semantic equivalence; likelihoods for weighted estimation	Sampling and clustering	Estimator access differs by variant
Probe / Lookback Lens	Attention maps and labeled training spans	Feature extraction and probe inference	Supervised contextual proxy, not proof of truth
Factuality alignment / FactAlign	Generator training and sentence-level factual feedback	Verification plus optimization	Reward errors can be reinforced
Retrieval control / Self-RAG	Training, retrieval, reflection-token supervision	Training and retrieved candidates	Reflection judgments need auditing
Contrastive decoding / CAD	Context-conditioned and context-free logits	Extra distribution at decoding	Greater source adherence can amplify false evidence
Layer contrast / DoLa	Intermediate layer predictions	Layer-output computation	Parametric knowledge is not external evidence
Representation editing / TruthX	Hidden states and learned auxiliary components	Auxiliary learning and intervention	Transfer beyond training conditions is uncertain
Verification / CoVe, CRITIC	Model calls; CRITIC also needs tools	Verification and revision rounds	Isolated prompts are not independent knowledge
Selective generation / SEAL	Rejection-token training and regularized decoding	Training; altered answering behavior	Lower risk must be interpreted at matched coverage

dently collected labels or an independently audited evaluator. Test natural errors from held-out generators, difficult consistent falsehoods, evidence conflicts, context permutations, paraphrases, and updated facts. Disaggregate performance rather than interpreting an average as proof of transfer. Verify with Caution and temporal benchmark analysis motivate independent error inspection and timestamped evidence (Godbole and Jia, 2025; Jiang et al., 2026).

Table 8 summarizes method access and deployment trade-offs. Report offline annotation and training costs separately from inference calls, sampled tokens, retrieval requests, tool invocations, latency, and memory. A black-box method can be expensive, while an internal method can be unavailable even if computationally economical. Do not infer a fixed latency multiplier from the number of model distributions or stages without measurement. Comparisons should either match budgets or present a quality–cost trade-off.